

Erin Zhang

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SUMMARY

Software Developer with experience owning a 50+ script Python/R automation codebase for insurance administration and actuarial reporting. Built production ETL workflows, Django-based internal tools, and pytest-backed calculation systems, reducing daily actuarial processing from 5+ hours of manual Excel work to approximately 3 seconds.

EDUCATION

University of Waterloo

Double Major in Mathematical Finance and Statistics, with Computing Science minor

Expected Dec 2026

Waterloo, Canada

- **President's Scholarship**
- Relevant Coursework: Algorithms & Data Structures, Machine Learning, Database Systems, Applications Software Engineering, Data-Intensive Distributed Analytics, Computational Mathematics
- Co-op Performance: Earned **Excellent or Outstanding** employer evaluations across all 6 co-op work terms.

HarvardX

CS50P: Programming with Python

Completed

100% grade

TECHNICAL SKILLS

Languages: Python, SQL, R, Bash, JavaScript

Frameworks & Libraries: Django, React, pandas, NumPy, pytest

Databases & Data Tools: MySQL, PostgreSQL, Microsoft SQL Server, SQLite, Tableau, Excel

Automation & Infrastructure: Git/GitHub, Linux, ETL pipelines, FTP/SFTP workflows, REST API design

Testing & Quality: Unit testing, integration testing, regression testing, data validation, test automation

Domain: Actuarial systems, financial data processing, insurance product modeling, quantitative trading

EXPERIENCE

Actuarial Software Developer (Python & R)

Atella Consulting

Jan 2026 – Aug 2026

North Carolina, United States

- Owned end-to-end development of a 50+ script Python/R automation codebase powering 10 recurring daily actuarial reports, reducing daily calculation runtime from 5+ hours of manual Excel-based processing to ~3 seconds of automated execution.
- Designed snapshot-based reporting infrastructure with timestamped directories, structured run metadata, and reproducible outputs — built on automated FTP/SFTP ingestion, DBF/CSV processing, and reference table joins to enable full traceability across daily and month-end cycles.
- Implemented modular calculation logic for complex insurance product rules — policy-level mapping, rate selection, interest accrual, effective-date allocation, and withdrawal/distribution handling — translating ambiguous domain requirements into deterministic, well-tested Python.
- Built full-stack features for a Django-based internal web application, connecting server-rendered review pages with backend calculation logic and automated reporting workflows; expanded QA coverage with pytest unit and regression tests to catch regressions across recurring production runs.

Data Reporting Analyst (SQL & Python)

Kenna Communications

Sep 2024 – Apr 2025

Mississauga, Canada

- Developed automated reporting pipelines in Python and SQL that consolidated data from three sources, eliminated 15 hours of manual work per week, and delivered recurring operational reports.
- Built a Python-based forecasting pipeline with automated validation testing on 50K+ records, achieving a 5% reduction in material waste (approx. \$150K annual savings).
- Designed optimized SQL queries using CTEs and window functions to power Tableau dashboards for real-time operational monitoring and shift-level performance tracking.

Data Research Analyst (R & Linux)

Chinese Academy of Sciences - Lanshi Institute of AI in Medicine

Jan 2024 – Apr 2024

Remote

- Developed automated QA pipelines for 10TB of genomic sequencing data using R, improving reproducibility and ensuring 98% downstream analysis accuracy.
- Applied machine learning methods to enhance cancer gene-sequencing platform predictive accuracy by 15%.

Quantitative Analyst Intern (Python)

AssetPro

Sep 2023 – Dec 2023

Shanghai, China

- Built a Python backtesting framework for factor-based trading signals on S&P 500 data, implementing walk-forward testing and out-of-sample evaluation.
- Reduced daily market-data processing time from 2 hours to under 15 minutes by parallelizing data pipelines with Python multiprocessing.

PROJECTS
